MOCK

NAME (pinyin):________________ NAME (汉字):_______

S&T2025 Chemistry Checkpoint 1 Mock Test [110 marks]

To simplify all the calculations (mock and actual tests), use integer atomic mass (exception: Cl – 35.5) and 0 ^{o}C = 273 K, but this sentence will not appear here on actual paper.

1. Write the formula / English name of the following substances. [10 marks]

Name	Formula
Acetic acid	
Hydrogen sulfide	
Carbon monoxide	
Potassium permanganate	
Boron bromide	
	SO_3
	PbI_2
	Na_2O_2
	$CaCO_3$
	$FeSO_4$

2. Write the English name of the element, full electronic configuration of the atom/ion and the number of unpaired electron(s) of the atom/ion. [10 marks]

Atom/ion	Name of element	Electronic configuration	No. of unpaired e^-
He			
Li^+			
Cl^+			
Au^{3+}			
Sg	No need to answer		

3. Write the English name of the elements: H ~ Zn, Br, Ag, Sn, I, Ba, W, Pt, Au, Hg, Pb, U. [10 marks]

4. Using VSEPR, draw the 3-D structure of the following molecules / ions. Indicate the orientations of the bonds and the lone pairs surrounding the central atom. State the geometry of each molecule/ion, with and without lone pair. [10 marks]

Structure	N_2H_4	SO_4^{2-}
Geometry including lone pair		
Geometry excluding lone pair		
Structure	BrF_3	SCN^-
Geometry including lone pair		
Geometry excluding lone pair		

5. Iodometric titration is a commonly used redox titration in the measurement of copper content in an unknown sample. The titrant used is sodium thiosulfate pentahydrate. Thiosulfate ion is obtained by replacement of one oxygen atom in sulfate ion with a sulfur atom. A 100.0 cm^3 standard solution is prepared from 1.2486 g sodium thiosulfate pentahydrate.

   (a) Write the formula of sodium thiosulfate pentahydrate and calculate the concentration of the standard solution. [2 marks]

(b) Write the ionic equation between thiosulfate and iodine. What is the color change? [2 marks]

A sample of copper ore (2.000 g) is dissolved in concentrated nitric acid. Assume the copper ore contains copper in oxidation states of 0 and +2.

(c) Write the ionic equations for the dissolution of the copper element. Use CuO to represent the copper in +2 oxidation state. [3 marks]

The nitrate ion in the obtained solution may interfere with the iodometric titration, because nitrate may oxidize thiosulfate under acidic condition, forming nitrogen monoxide gas. Thiosulfate forms the same product as question (b).

(d) Write the ionic equation of this redox reaction. [2 marks]

In order to remove the interference of nitrate, excess sodium hydroxide solution is added. The blue precipitate is collected by technique **X** and re-dissolved in sulfuric acid. The final volume of the copper solution **Y** is 250.0 cm^3.

(e) Write the formula of the precipitate and the name of technique **X**. [2 mark]

20.00 cm^3 of solution **Y** is pipetted into a 250 cm^3 conical flask. 1.0 g potassium iodide solid is added and mixed with the solution. The color of the solution turns brown with formation of a white precipitate.

(f) Given the precipitate is a binary compound with copper(I), write the ionic equation. [2 marks]

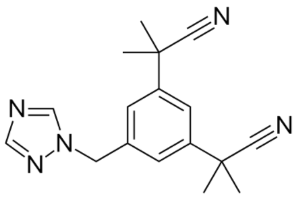
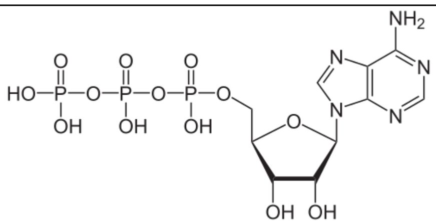
The content in the conical flask is <u>immediately</u> titrated with sodium thiosulfate solution until the color of the solution turns light yellow. A few drops of solution **Z** are added and the solution in the conical flask turns blue. Titration continues until the solution becomes colorless. The total volume of thiosulfate used is 19.79 cm^3.

(g) Why is it necessary to conduct the titration <u>immediately</u>? [1 mark]

(h) Write the name of solution **Z**. [1 mark]

(i) Calculate the mass percentage of copper element in the copper ore. Report your results with four significant figures. [5 marks]

6. Count the number of sp^3, sp^2 and sp carbons in each molecule. [10 marks]

Molecule	Number of **Carbon** atoms		
	sp^3	sp^2	sp
			
			

7. 8.00 g of ionic compound **X** was put into a 1 dm^3 closed vessel and air was then removed from the vessel. When heated at 250 ^oC, **X** completely decomposed to two gases (**Y** and **Z**) without any solid residue. The pressure in the vessel was 13.05 bar. When the vessel was cooled down to 25 ^oC, the pressure decreased to 2.48 bar.

(a) Calculate the molar ratio of the two gases formed after the decomposition. [3 marks]

(b) What is the formula of one of the two gases, **Y**? [1 mark]

(c) Calculate the molar mass of the other gas **Z**. [3 marks]

(d) Deduce the formula of the ionic compound **X**. [3 marks]

8. The rate constant of a reaction is $k_1 = 2.50\times10^{-3}$ mol dm^{-3} s^{-1} at 25 °C and $k_2 = 5.00\times10^{-3}$ mol dm^{-3} s^{-1} at 50 °C.

(a) Express the rate constants in the unit of mol dm^{-3} h^{-1}. [2 marks]

(b) Calculate the activation energy. [5 marks]

(c) Calculate the rate constant at 75 ^{o}C. [3 marks]

9. Solution **A** is prepared by dissolving 3.375 g anhydrous sodium acetate in water and the total mass of the solution is 100.000 g. K_b(water) = 0.512 K kg mol^{-1} and K_f(water) = 1.86 K kg mol^{-1}.

(a) Calculate the molality of solution **A**. [3 marks]

(b) Calculate the boiling point and the melting point of solution **A**. Report your results with 3 decimal places. [6 marks]

(c) An acetic acid solution **B** with the same molality (as solution **A**) is prepared, from 1.500 mol dm^{-3} acetic acid solution **C** (density: 1.02 g cm^{-3}) and deionized water. Given the mass of solution **B** is 100.000 g, calculate the mass of deionized water and solution **C**. If you are unable to find the molality in question (a), use 0.1000 mol kg^{-1}. [6 marks]

(d) Given that the boiling point of solution **B** is 100.229 ^{o}C, calculate the van't Hoff factor of solution **B**. Explain why the factor is between 1 and 2. [5 marks]

Periodic Table of Elements with Relative Atomic Masses

Use integer atomic mass (exception Cl) for calculation. 0 ^{o}C = 273 K.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc [98]	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra 226.0	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (269)	107 Bh (270)	108 Hs (270)	109 Mt (278)	110 Ds (281)	111 Rg (282)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (290)	116 Lv (293)	117 Ts (294)	118 Og (294)

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (254)	100 Fm (257)	101 Md (256)	102 No (254)	103 Lr (257)

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